IROHUB

Python Syllabus Advanced

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Overview & Environment Setup

- Introduction to Python Programming
- Features of Python
- Application Fields
- Installation

Basics of Programming

- Identifiers and Variables
- Value assignments
- Introduction to Datatypes
- User Inputs
- Comments

Operators and Types

- Introduction to Operators
- Arithmetic Operators
- Relational Operators
- Assignment Operators
- Logical Operators
- Bitwise Operators
- Membership Operators
- Identity Operators

Datatypes

- Introduction to Datatypes and Type Conversions
  - Basic Datatypes
    - Numbers
    - Strings
    - Boolean Datatype
  - Advanced Datatypes
    - Lists
    - Tuples
    - Dictionaries
    - Sets
  - Comprehensions
    - Dictionary Comprehensions
    - Set Comprehensions
    - List Comprehensions

Conditional Statements

- Importance of Conditional Statements
- Implementing if, if-else and if-elif-else statements

Looping Statements

- Introduction to Loops
- While Loops
- For Loops
  - Flow Control Statement
    - Break
    - Continue
    - Else
    - Pass

Functions

- Importance of Functions
- Understanding Built-In Functions
- User-Defined Functions
- Passing Arguments to a Function
- Types of Argument passes
- Local and Global Variables
- Recursive Functions
- Lambda Functions
- Generator Functions and Yield Statements
- Map, Filter and Reduce Functions

Modules

- Importing Modules
- Built-In Modules
- User-defined Modules

Object-Oriented Programming Concepts

- Creating Classes and Objects
- Class Instantiation
- Constructors and Destructors
- Super Functions
- Method Resolution Order (MRO)
- Data Mangling
- Inheritance
- Polymorphism
- Data Abstraction
- Encapsulation principles

Exception Handling

- Exceptions and Types
- Understanding the Exception Base Class
- Handling Exceptions
- Raising Custom Exceptions
- User Defined Exception Classes

File Handling

- Understanding File Handling Principles
- Opening a File
- Reading from Files
- Writing to a File
- Closing Files
- Pathlib
- OS and Shutil Modules
- Context Managers - "with" - for File Operations

Web Development from end technologies (basics)

- HTML
- CSS
- JavaScript
- Bootstrap

Python Advanced Libraries

- Numpy
- Pandas
- Regex - Regular Expressions
- Tkinter
- Opencv

Database Management Systems

- Understanding SQL and NoSQL
- SQL Queries
- Operators
- Subqueries
- Filtering Queries
- Aggregate Functions
- Joins and Types of SQL Joins
- SQL Clauses
- ERM Concept

Web Frameworks

- Understanding the Django Web Framework
- MVT and MVC Architectures
- URL Routing
- View Handling
- Template Rendering
- Model Creation and setting Field Types
- Migration Policies
- ORM Querying
- DTL - Django Templating Language
- Database Connection Managements
- Working with Static and Media Files
- Introduction to Flask Framework
- Routing in Flask
- Views and Controllers
- Configuration Steps
- Flask Extensions
- Database Integration

Git and Github

- Walkaround on Git and Github Environments
- Importance of Version Control Techniques
- Repository Creation
- Pushing code to Github
- Hosting to pythonanywhere.com from Github

Angular Framework

- Introducing Angular Frame Work
- Introducing Angular Typescript
- Difference BT Angular And Angular JS
- Basic Functions And Types Of Arguments
- oops Concepts
- Introducing Angular Projects
  - Binding and types of binding
    - Style binding
    - Property binding
    - Event Binding
- Angular Structural Directieves

APIs & Django Rest frame work

- Introducing API
- API different http methods
- Introducing Serializers and Model Ster
- Introducing Postman an usage
  - Various methods For CRUD Operations
    - Function Base API Views(FBV)
    - RESTAPI (Restframework)
    - Class Base API Views(CBV)
- Difference between FBV and CBV

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Student Benefits

- Project to practice
- Practical classes
- Resume Preparation support
- Mock Interviews
- Technical round and machine test

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